Aggregating Methods

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DOBI–Dombi Bonferroni Method

The DOBI method, which stands for Dombi Bonferroni, is a technique used in data analytics for aggregating intuitionistic fuzzy numbers (IFNs). This method combines the Bonferroni mean (BM) with Dombi operators, which are advantageous for considering the interrelationships among parameters and capable of handling fuzzy information.
This method has the following features:

- particularly useful in multi-attribute group decision making;
- handles fuzzy infornation.

How DOBI works

Here are some key points:

- IFS;
- Dombi t-Norm and t-Conorm;
- Dombi Operations;
- Bonferroni Mean Operator;
- Combine Dombi and Bonferroni.

About the IFS and the IFNs

The following definitions and concepts form the basics of IFNs, which play an important role in MCDM.

- Intuitionistic Fuzzy Set (IFS)
- Interval-Valued Intuitionistic Fuzzy Set (IVIFS)
- Intuitionistic Fuzzy Number (IFN)
- Membership and Non-Membership
- Hesitation Degree
- Intuitionistic Fuzzy Interval Number (IFIN)

In short, an IFN is characterized by a membership function μ and a non-membership function υ, where μ represents the degree of membership and υ represents the degree of non-membership.

About the IFS and the IFNs

Definition. (IFS)

Let X be a nonempty set. An intuitionistic fuzzy set on X is given by: $A = \{\langle x, \mu_A(x), \upsilon_A(x)\rangle | x \in X\}$, where $\mu_A(x)$ denotes the degree of membership and $\upsilon_A(x)$ denotes the degree of non-membership of the element $x \in X$ to the set A, satisfying the condition that $0 \leq \mu_A(x) + \upsilon_A(x) \leq 1, \forall x \in X$.

Note that $\mu_A(x)$ and $\upsilon_A(x)$ are often given by experts or some algorithm. For the element $x, \langle \mu_A(x), \upsilon_A(x)\rangle$ is called intuitionistic fuzzy numbers (IFNs), and for convenience, it is written as $\tilde{a} = (\mu_a, \upsilon_a)$.

Definition. (Hesitancy Degree)

If $\forall x \in X$, $\pi(x) = 1 - \mu_A(x) - \upsilon_A(x)$, then $\pi(x)$ is called the hesitancy degree of x to X. It is obvious that $0 \leq \pi(x) \leq 1$.

Extending the Concept to Pythagorean Fuzzy Set (PFS)

Definition. (PFS)

Let X be a nonempty set. A PFS on X is given by: $P = \{\langle x, \mu_P(x), \upsilon_P(x)\rangle | x \in X\}$, satisfying the condition that $0 \leq (\mu_P(x))^2 + (\upsilon_P(x))^2 \leq 1, \forall x \in X$.

For the element $x, (\mu_P(x), \upsilon_P(x))$ is called a PFN, and for convenience, a PFN is denoted by $\tilde{p} = (\mu_{\tilde{p}}, \upsilon_{\tilde{p}})$.

Definition. (score function)

$$S(\tilde{p}) = \frac{1}{2}(1 + (\mu_{\tilde{p}})^2 - (\upsilon_{\tilde{p}})^2), \quad S(\tilde{p}) \in [0, 1] \tag{1}$$

Definition. (accuracy function)

$$A(\tilde{p}) = (\mu_{\tilde{p}})^2 + (\upsilon_{\tilde{p}})^2, \quad A(\tilde{p}) \in [0, 1] \tag{2}$$

Bonferroni Mean Operator

Definition. Bonferroni Mean (BM)

Let $p, q \geq 0$ and $a_i, (i = 1, 2, \cdots, n)$ be a collection of non-negative numbers. Then the operation

$$BM^{p,q}(a_1, a_2, \cdots, a_n) = \left(\frac{1}{n(n-1)} \sum_{i,j=1}^{n} a_i^p a_j^q\right)^{\frac{1}{p+q}} \quad (i \neq j) \tag{3}$$

is called BM.

Definition. Geometric Bonferroni Mean (GBM)

$$GBM^{p,q}(a_1, a_2, \cdots, a_n) = \frac{1}{p+q}\left(\prod_{i,j=1}^{n} (pa_i + qa_j)\right)^{\frac{1}{n(n-1)}} \quad (i \neq j) \tag{4}$$

Dombi t-Norm and t-Conorm

Definition. (t-norm)

A function T : $[0,1] \times [0,1] \to [0,1]$ is called a t-norm if it meets the following conditions:
(1) $T(1,x) = x, \forall x$;
(2) $T(x,y) = T(y,x), \forall x, y$;
(3) $T(x, T(y,z)) = T(T(x,y),z), \forall x, y, z$;
(4) If $x_1 \leq x_2$ and $y_1 \leq y_2$ then $T(x_1, y_1) \leq T(x_2, y_2)$.

Definition. (t-conorm)

A function S : $[0,1] \times [0,1] \to [0,1]$ is called a t-norm if it meets the following conditions:
(1) $S(0,x) = x, \forall x$;
(2) $S(x,y) = S(y,x), \forall x, y$;
(3) $S(x, S(y,z)) = S(S(x,y),z), \forall x, y, z$;
(4) If $x_1 \leq x_2$ and $y_1 \leq y_2$ then $S(x_1, y_1) \leq S(x_2, y_2)$.

Dombi t-Norm and t-Conorm

Dombi operations, which will be defined later, are generated through Dombi t-Norm ($\Delta(\mu, \upsilon)$) and t-Conorm ($\Delta_C(\mu, \upsilon)$).

$$\Delta(\mu, \upsilon) = \frac{1}{1 + ((\frac{1-\mu}{\mu})^r + (\frac{1-\upsilon}{\upsilon})^r)^{\frac{1}{r}}} \tag{5}$$

$$\Delta_C(\mu, \upsilon) = 1 - \frac{1}{1 + ((\frac{\mu}{1-\mu})^r + (\frac{\upsilon}{1-\upsilon})^r)^{\frac{1}{r}}} \tag{6}$$

where $r > 0$ and $\mu, \upsilon \in [0, 1]$

Dombi Operations (of IFNs)

Let $\tilde{a} = (\mu_a, \upsilon_a)$ and $\tilde{b} = (\mu_b, \upsilon_b)$ be two IFNs, and $\lambda \geq 0$. Then the Dombi Operations can be defined as follows:

Addition "$\oplus$"

$$\tilde{a} \oplus \tilde{b} = (\mu_a + \mu_b - \mu_a\mu_b, \upsilon_a\upsilon_b) \tag{7}$$

Multiplication "$\otimes$"

$$\tilde{a} \otimes \tilde{b} = (\mu_a\mu_b, \upsilon_a + \upsilon_b - \upsilon_a\upsilon_b) \tag{8}$$

Scalar multiplication "$\cdot$"

$$\lambda \cdot \tilde{a} = (1 - (1 - \mu_a)^\lambda, (\upsilon_a)^\lambda) \tag{9}$$

Power

$$\tilde{a}^\lambda = ((\upsilon_a)^\lambda, 1 - (1 - \upsilon_a)^\lambda) \tag{10}$$

Dombi Operations (of PFNs)

Let $\tilde{p}_1 = (\mu_1, v_1)$ and $\tilde{p}_2 = (\mu_2, v_2)$ be two PFNs, and $\lambda \geq 0$. $\tilde{\mu}_i := \frac{\mu_i^2}{1-\mu_i^2}$, $\tilde{v}_i := \frac{v_i^2}{1-v_i^2}$. Then the Dombi Operations can be defined as follows:

Addition "$\oplus$"

$$\tilde{p}_1 \oplus \tilde{p}_2 = \left(\sqrt{1 - \frac{1}{1 + ((\tilde{\mu_1})^r + (\tilde{\mu_2})^r)^{\frac{1}{r}}}}, \frac{1}{\sqrt{1 + ((\tilde{v_1})^{-r} + (\tilde{v_2})^{-r})^{\frac{1}{r}}}}\right) \quad (11)$$

Multiplication "$\otimes$"

$$\tilde{p}_1 \otimes \tilde{p}_2 = \left(\frac{1}{\sqrt{1 + ((\tilde{\mu_1})^{-r} + (\tilde{\mu_2})^{-r})^{\frac{1}{r}}}}, \sqrt{1 - \frac{1}{1 + ((\tilde{v_1})^r + (\tilde{v_2})^r)^{\frac{1}{r}}}}\right) \quad (12)$$

Dombi Operations (of PFNs)

Scalar multiplication ”·”

$$\lambda \cdot \tilde{p} = \left(\sqrt{1 - \frac{1}{1 + (\lambda \tilde{\mu}^r)^{\frac{1}{r}}}}, \frac{1}{\sqrt{1 + (\lambda \tilde{\upsilon}^{-r})^{\frac{1}{r}}}}\right) \tag{13}$$

Power

$$\tilde{p}^{\lambda} = \left(\frac{1}{\sqrt{1 + (\lambda \tilde{\mu}^{-r})^{\frac{1}{r}}}}, \sqrt{1 - \frac{1}{1 + (\lambda \tilde{\upsilon}^r)^{\frac{1}{r}}}}\right) \tag{14}$$

Honestly speaking I don't quite understand the DOBI method. What confuse me:

- where the fuzzy set comes from
- why we have to define so many operators
- the difference between IFS and PFS

Plus, I couldn't find the original article where the DOBI method was proposed. I searched for Dombi operators and Bonferroni means separately, but still didn't find satisfying results.

ARLON–Alternative Ranking using two-step LOgarithmic Normalization

Instead of a single normalization process, two-step normalization processses are executed to enhance the robustness of normalization values and determine the most effective solution for the decision problem.
The ARLON method has the following features:

- diverse findings resulting from different logarithm normalizations;
- investigation into the impacts of changes in the significance levels of normalizations;
- utilization of macro-level data in performance evaluation.

How ARLON works

Comprising four steps, ARLON is structured as follows:

- the initial decision matrix
- two-step logarithmic normalization
- construct the weighted aggregated normalized matrix
- the final ranking of alternatives

The initial decision matrix with respect to input data is formulated using:

$$X = \begin{bmatrix} x_{11} & \dots & x_{1j} & \dots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \dots & x_{ij} & \dots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{m1} & \dots & x_{mj} & \dots & x_{mn} \end{bmatrix}_{m\times n} .$$

In this decision matrix, x_{ij} denotes the assesment of the ith alternative in jth criterion.

Two-step Logarithmic Normalization

The first logarithmic normalization is computed using the following equations, where $\mathbb{B}$ stands for benefit criteria and $\mathbb{C}$ stands for cost criteria.

$$\mathbb{N}_{ij}^{+} = \frac{\ln(x_{ij})}{\ln(\prod_{i=1}^{m} x_{ij})}, \quad i = 1, 2, \cdots, m; \quad j \in \mathbb{B}, \tag{15}$$

$$\mathbb{N}_{ij}^{-} = \frac{1 - \frac{\ln(x_{ij})}{\ln(\prod_{i=1}^{m} x_{ij})}}{m-1}, \quad i = 1, 2, \cdots, m; \quad j \in \mathbb{C}. \tag{16}$$

The second logarithmic normalization is calculated as follows:

$$\aleph_{ij}^{+} = \frac{\log_2(x_{ij})}{\sum_{i=1}^{m} \log_2(x_{ij})}, \quad i = 1, 2, \cdots, m; \quad j \in \mathbb{B}, \tag{17}$$

$$\aleph_{ij}^{-} = 1 - \frac{\log_2(x_{ij})}{\sum_{i=1}^{m} \log_2(x_{ij})}, \quad i = 1, 2, \cdots, m; \quad j \in \mathbb{C}. \tag{18}$$

Two-step Logarithmic Normalization

Then aggregate the two values by using Heron mean aggregation operator.

Definition. (Heron Mean)

$$h_{ij}(\mathbb{N}_{ij}, \aleph_{ij}) = (1-r)\sqrt{\mathbb{N}_{ij}\aleph_{ij}} + r\frac{\mathbb{N}_{ij} + \aleph_{ij}}{2} \tag{19}$$

$r \in [0, 1]$ illustrates the impact of each normalization value, providing flexibility to the decision maker.

Ultimately the aggregated normalized decision matrix is obtained.

$$\begin{bmatrix} h_{11} & \dots & h_{1j} & \dots & h_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{i1} & \dots & h_{ij} & \dots & h_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{m1} & \dots & h_{mj} & \dots & h_{mn} \end{bmatrix}_{m\times n}.$$

Construct the weighted aggregated normalized matrix

The normalized values are multiplied by the criterion weights to generate the weighted matrix.

$$\hbar_{ij} = w_j \cdot h_{ij} \tag{20}$$

As a result, the weighted aggregated normalized matrix is obtained.

$$H = \begin{bmatrix} \hbar_{11} & \dots & \hbar_{1j} & \dots & \hbar_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \hbar_{i1} & \dots & \hbar_{ij} & \dots & \hbar_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \hbar_{m1} & \dots & \hbar_{mj} & \dots & \hbar_{mn} \end{bmatrix}_{m \times n} .$$

The Final Ranking of Alternatives

The weighted aggregated normalized values of benefit criteria and cost criteria are separately summed:

$$\mathbb{B}_i = \sum_{j \in \mathbb{B}} \hbar_{ij} \quad ; \tag{21}$$

$$\mathbb{C}_i = \sum_{j \in \mathbb{C}} \hbar_{ij} \quad . \tag{22}$$

The final ranking is determined using:

$$\mathbb{S}_i = \mathbb{B}_i^{\psi} + \mathbb{C}_i^{1-\psi} \quad , \tag{23}$$

where the ψ value is calculated as the ratio of the total number of benefit criteria influencing the alternative ranking to the overall number of criteria.

VIKOR– the multi criteria compromise ranking method

Its features:

- based on LP-metric
- the closest alternative to the ideal solution is preferred

The Decision Matrix

$$X = \begin{bmatrix} x_{11} & \dots & x_{1j} & \dots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \dots & x_{ij} & \dots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{m1} & \dots & x_{mj} & \dots & x_{mn} \end{bmatrix}_{m\times n} .$$

In this decision matrix, x_{ij} denotes the assesment of the ith alternative in jth criterion.

the LP-metric

This is a compromising method, which uses the LP-metric method to find the closest alternative to the optimal solution.

$$L_{pi} = (\sum_{j=1}^{n} (\frac{w_j(f_j^* - f_{ij})}{f_j^* - f_j^-})^p)^{\frac{1}{p}}; \quad i = 1, \cdots, m, 1 \leq p \leq \infty \tag{24}$$

where w_j indicates the weight of attribute declared by the decision maker, $f_{ij} = x_{ij}$, f_j^* is the best and f_j^- is the worst.

L_{1i} is introduced by s_i and $L_{\infty i}$ is introduced by R_i

$$S_i = \sum_{j=1}^{n} w_j \frac{f_j^* - f_{ij}}{f_j^* - f_j^-} \tag{25}$$

$$R_i = \max_j (w_j \frac{f_j^* - f_{ij}}{f_j^* - f_j^-}) \tag{26}$$

f_j^* and f_j^-

For positive attributes:

$$\begin{cases} f_j^* = \max_i f_{ij} \\ f_j^- = \min_i f_{ij} \end{cases} \tag{27}$$

For negative attributes:

$$\begin{cases} f_j^* = \min_i f_{ij} \\ f_j^- = \max_i f_{ij} \end{cases} \tag{28}$$

The Final Ranking

The VIKOR Index

The VIKOR index is calculated for each alternative:

$$Q_i = \nu \frac{S_i - S^*}{S^- - S^*} + (1 - \nu) \frac{R_i - R^*}{R^- - R^*} \tag{29}$$

where ν indicates the strategic weight, which is often considered equal to 0.5.

During this step, the alternatives are ranked as descending in values of (S) and (R) and (Q). The alternative with the lowest amount in attributes is the superior alternative.

The End